



Koneru Lakshmaiah Education Foundation

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OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026-27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.: 10

Team No.: 18

Project Title: LINUX FILE SYSTEM EXPLORER USING SYSTEM CALLS

Team Members

Roll Number	Student Name	Signature
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2520090161	Koumudi	<i>Koumudi</i>
2520090073	Mrudula	<i>Mrudula</i>

1. Abstract

The Linux File System Explorer Using System Calls is a command-line based system programming project designed to provide users with an interactive way to explore and manage files and directories in a Linux environment. The project addresses the limitations of relying only on conventional graphical file managers by demonstrating how operating-system-level file management can be implemented using Linux/POSIX interfaces. The system provides essential operations such as viewing directory contents, navigating directories, creating and deleting files and directories, reading and writing files, renaming files, searching for files, and displaying file metadata such as size, permissions, ownership, and modification time. The proposed approach is to develop the application in the C programming language with a modular command-line interface. Each operation is implemented using appropriate Linux/POSIX system calls and library interfaces, with proper error handling and validation to ensure safe and reliable execution. The project makes use of interfaces such as `open()`, `read()`, `write()`, `close()`, `mkdir()`, `rmdir()`, `unlink()`, `rename()`, `stat()`, `chdir()`, `getcwd()`, `opendir()`, `readdir()`, and `closedir()`. Through these technologies, the project demonstrates important concepts of file descriptors, file I/O, directory management, metadata handling, permissions, and system-level programming. The expected outcome is a functional and user-friendly Linux file system explorer that performs common file-management tasks directly through system-level interfaces while providing practical understanding of how Linux manages files and directories.

2. Problem Statement

Managing files and directories is a fundamental operation in any operating system. Although graphical file managers provide convenient interfaces, they often hide the underlying system-level mechanisms used to perform file and directory operations. Students learning Linux system programming therefore need a practical application that demonstrates how these operations can be performed directly through Linux/POSIX interfaces. The proposed project solves this problem by developing a command-line Linux File System Explorer that allows users to perform common file and directory operations while explicitly demonstrating the system calls and APIs responsible for those operations.

3. Objectives

1. To develop a command-line system explorer for Linux.
2. To demonstrate practical use of Linux/POSIX system calls and file-management API's.
3. To implement file and directory creation, deletion, renaming, reading, writing, and navigation.
4. To display important file metadata such as size, permissions, ownership, and modification time.
5. To provide a file-search facility within selected directories.
6. To implement proper error handling for invalid paths, permissions, and unavailable files.
7. To improve understanding of file descriptors, directory handling, system calls, and Linux file-system concepts.

4. Proposed Methodology

The project will be implemented in C on a Linux environment using a menu-driven or command-based terminal interface. First, the application will obtain and display the current working directory and allow the user to navigate through directories. Directory contents will be retrieved and displayed using directory-handling interfaces. File operations such as creation, reading, writing, and deletion will be implemented using appropriate file descriptors and Linux/POSIX interfaces. The `stat()` interface will be used to obtain file metadata, while directory operations will use interfaces such as `opendir()` and `readdir()`. User inputs will be validated before operations are performed, and system errors will be handled using return values and suitable error-reporting mechanisms. The application will be organized into separate functional modules so that file management, directory navigation, metadata display, and search operations can be developed and tested independently. Finally, all modules will be integrated into a single command-line file system explorer and tested with different files, directories, permissions, and invalid inputs.

5. Operating Systems Concepts / Linux APIs Used

OS Concept / Linux API / System Call	Purpose in the Project
<code>open()</code> , <code>read()</code> , <code>write()</code> , <code>close()</code>	Create/open, read, write, and close files using file descriptors

mkdir(), rmdir(), unlink(), rename()	Create, delete, and rename files and directories
stat()	Display file metadata such as size, permissions, ownership, and modification time.
chdir(), getcwd()	Navigate between directories and display the current working directory
opendir(), readdir(), closedir()	Open directories and display the current working directory

6. Individual Contribution

Roll Number	Student Name	Individual Responsibility
2520090161	Koumudi	File operations using open(), read(), write(), unlink(), and rename()
2520090049	Usha Rani	Directory navigation and management using mkdir(), rmdir(), chdir(), getcwd(), opendir(), and readdir()
2520090073	Mrudula	File metadata, search functionality, error handling, testing, and project integration.

7. Tools / Platforms / Software Used

Tool / Platform / Software	Purpose
Linux / Ubuntu	Operating system environment for developing and running the project
C / C++	Main programming language used to implement the file system explorer
GCC Compiler	Compiling the C source code
Terminal	Running and interacting with the command-line application
POSIX/ Linux APIs	Performing file, directory, navigation and metadata operations

8. Expected Outcome

The expected outcome is a working command-line Linux File System Explorer that enables users to explore and manage files and directories through a simple terminal interface. The system will demonstrate practical use of Linux/POSIX system calls for file I/O, directory management, navigation, metadata retrieval, and file manipulation. It will provide meaningful error messages for invalid operations and permissions-related failures. The completed project will help users understand how high-level file-management operations interact with the Linux operating system and will provide hands-on experience with system calls, file descriptors, directory streams, file metadata, and POSIX programming concepts.

9. GitHub Repository

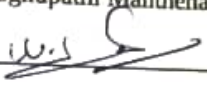
GitHub Repository Link: https://github.com/usharani-28/ossproject_2520090049/tree/main/KLH-CSIT-2nd%20year-18-19

Repository Created: yes

☐ Yes ☐ No

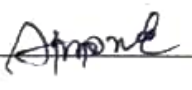
Faculty Name and Signature

Faculty Name: Raghupathi Manthana

Signature: 

Remarks: _____

Faculty Use

Project Approved: 

☒ Yes ☐ No

Date: 10/09/2020

Faculty Signature: 